

# List and Dictionary Comprehensions

## Project Title

### List and Dictionary Comprehensions

## Description

This project demonstrates the use of **List Comprehensions** and **Dictionary Comprehensions** in Python.

Comprehensions provide a concise and efficient way to create collections from existing data. In this project, a simple **Product Dataset** is used to demonstrate filtering, transformation, and collection creation.

## Objective

The main objectives of this project are:

- To understand List Comprehensions.
- To understand Dictionary Comprehensions.
- To create collections efficiently using Python.
- To perform filtering using conditions.
- To perform data transformation using comprehensions.
- To compare comprehensions with traditional Python loops.
- To understand how comprehensions can be useful in data preprocessing.

## Tools and Technologies

- Python
- Jupyter Notebook

## Dataset

A sample **Product Dataset** is used in this project.

The dataset contains the following information:

| Column                | Description       |
|-----------------------|-------------------|
| <code>id</code>       | Unique product ID |
| <code>name</code>     | Product name      |
| <code>category</code> | Product category  |
| <code>price</code>    | Product price     |

Example products include Laptop, Mouse, Keyboard, Chair, Table, Notebook, Pen, and Headphones.

## ◆ List Comprehension

List comprehension is a concise way to create a list from an iterable.

### Syntax

```
[expression for item in iterable]
```

### Example

```
product_names = [product["name"] for product in products]
```

This creates a list containing the names of all products.

## ◆ Dictionary Comprehension

Dictionary comprehension is used to create dictionaries in a concise way.

### Syntax

```
{key: value for item in iterable}
```

### Example

```
product_price = {  
    product["name"]: product["price"]  
    for product in products  
}
```

This creates a dictionary containing product names as keys and prices as values.

## Examples Included

### List Comprehension Examples

1. Extracting product names.
2. Extracting product prices.
3. Converting product names to uppercase.
4. Filtering products based on price.
5. Filtering products by category.
6. Applying a discount to product prices.

### Dictionary Comprehension Examples

1. Creating a product-price dictionary.
2. Creating a product-category dictionary.

3. Creating a dictionary containing only expensive products.

## Filtering

Conditions can be added to comprehensions using the `if` keyword.

Example:

```
expensive_products = [  
    product["name"]  
    for product in products  
    if product["price"] > 2000  
]
```

This selects only products whose price is greater than ₹2000.

## Transformation

Comprehensions can also be used to transform existing data.

Example:

```
uppercase_names = [  
    product["name"].upper()  
    for product in products  
]
```

This converts all product names into uppercase letters.

## Comprehension vs Traditional Loop

### Traditional Loop

```
names = []  
  
for product in products:  
    names.append(product["name"])
```

### List Comprehension

```
names = [product["name"] for product in products]
```

List comprehension provides the same result using fewer lines of code.

## Project Structure

```
List-Dictionary-Comprehensions/  
|  
├── List_Dictionary_Comprehensions.ipynb  
└── README.md
```

## Deliverables

- Jupyter Notebook containing List Comprehension examples.
- Jupyter Notebook containing Dictionary Comprehension examples.
- Filtering and transformation examples.
- Comparison between traditional loops and comprehensions.
- README documentation.

## Learning Outcome

After completing this project, I gained an understanding of how List and Dictionary Comprehensions can be used to write concise and readable Python programs.

I also learned how comprehensions can be applied to **filter, transform, and preprocess data**, which is useful in Data Science and Machine Learning tasks.

## Author

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